

Converting

Exercise n° 2

On the *Prelude n° 4 in E minor, Op. 28*, by Frédéric Chopin (1839) — public domain.

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What you will do: take a **MIDI** file of a well-known piece, convert it to the Music Motion format and open it in the Score, so you can hear it in Pythagorean tuning, see it written out, and edit or print it.

How long it takes: about twenty minutes. **You do not need to read music:** it is all clicks. **What you need:** the files that come with this lesson (`Preludio N° 4 Op28 - E min.mid` and `.mxl`), with the PDF of the score: [↓ download them here](#)) and a computer with a browser and sound.

Along the way you will understand something that shows up in everything else: why a MIDI file and a MusicXML file are not the same, even if they sound alike.

Three extensions and one word

.mid **MIDI.** It stores which key is played and when. What it does **not** store is *which note* it is: it writes «key 61», and key 61 can be a C# or a Db. It is the format you find most often on the internet.

.xml · .mxl **MusicXML.** The real score: besides the key, it stores how each note is *written* — whether it carries a sharp or a flat. The **.mxl** is the same file, compressed.

.mmf **The Music Motion format.** It takes that information and gives every note its exact pitch. It is the file the Keyboard, the Score and the Console open.

The copyist is the **PLAYER:** the app that reads a MIDI file and turns it into MMF.

THE TWO PATHS

The same piece can come into the Score through two doors. This exercise takes the first one, which is the one you will need almost every time, and at the end compares it with the second.

Path A — from a MIDI file

.mid → PLAYER → .mmf → SCORE

This lesson's path. It works for any MIDI file you find, and it is the normal case.

The copyist **works out** the spelling of each note from the key signature in the file, because the MIDI file does not carry it.

Path B — from a MusicXML file

.xml / .mxl → SCORE

Shorter: the Score opens it directly, without going through the Player.

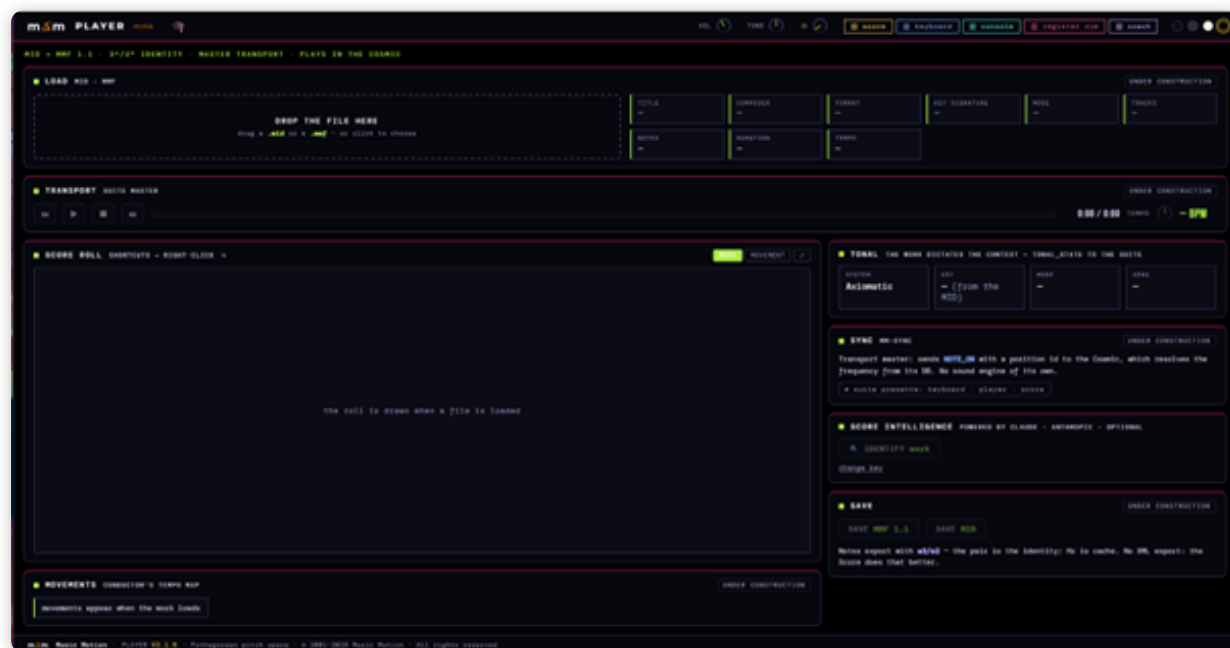
And better, when it exists: the spelling is **not worked out**, it is written in the file.

We take path A on purpose. Not because it is the best, but because it is the one you will run into: the internet has MIDI files of almost everything and MusicXML of very little. At the end of the exercise we open the same Prelude through path B and compare the two. That is where you see the difference with your own eyes.

PATH A — FROM A MIDI FILE TO YOUR SCORE

1 • Open the *PLAYER*

From the **COSMIC KEYBOARD**, the  **player** button on the navigation bar. Leave the Cosmic open: it makes the sound for the whole suite, and the Player uses it to play.



2 • Load the *MIDI* file

Two ways, whichever you prefer:

- **Drag** the file  and drop it on the Player's drop zone.
- Or **click** that zone and choose it in the file browser.

TIP: the Player accepts `.mid`, `.midi` and `.mmf`. If you pick an `.xml` nothing will happen: that one takes path B, straight into the Score.

Check: you should see the file name and, below it, the piece drawn as a grid of bars — each bar is a note, and the higher it sits, the higher the note. If nothing appears, the file was not a valid MIDI file.



3 • Listen to it

The button on the transport. The Player sends and the Cosmic sounds: you are hearing the Prelude in Pythagorean tuning, with pure fifths.

Check: if there is no sound, it is almost always one of two things: the **COSMIC KEYBOARD** was closed, or its volume is down. Open it again and try once more.

4 • The work card (optional)

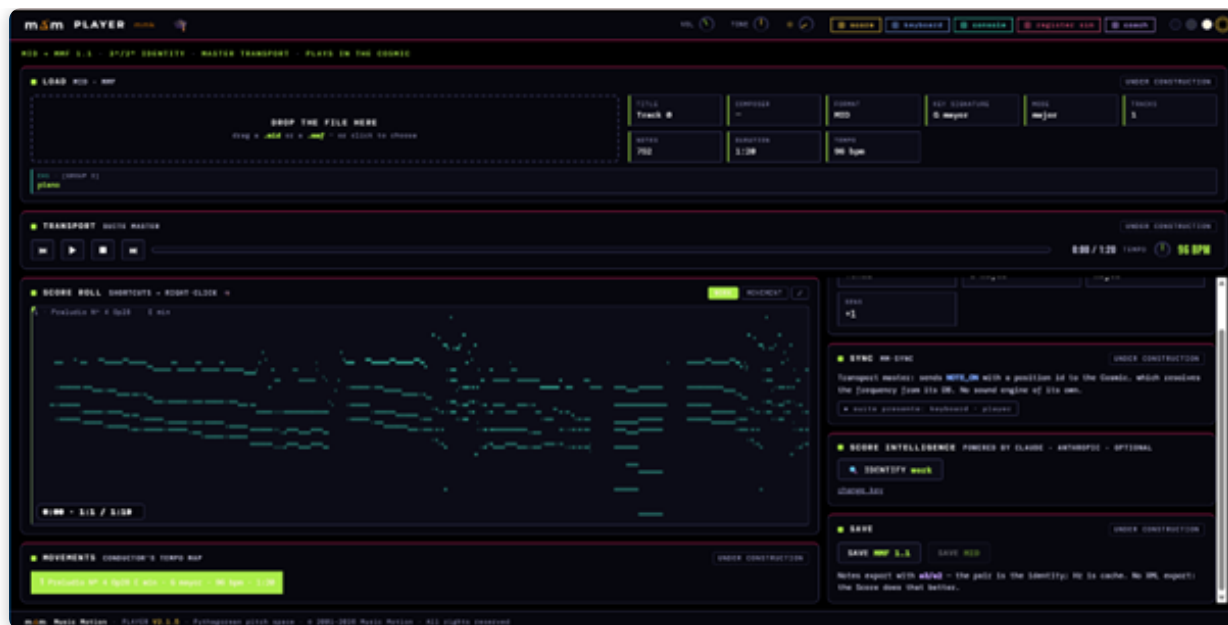
The IDENTIFY work button is **Score Intelligence**: it recognises which piece it is and what its tempos are, and fills in the `work` and `movement` fields.

This step needs a key of your own and an internet connection. If you do not have one, choose `not now` when it asks: **the exercise carries on the same** and nothing that follows depends on it. The `work` and `movement` fields can be typed in by hand.

5 • Save it as MMF

The `SAVE MMF 1.1` button. The file goes to your downloads folder, named after the piece.

Check: look in Downloads for a file ending in `.mmf`. That is the Prelude, now in the house format. You have just converted it.



6 • Open it in the SCORE

1. Open the **SCORE** — from the Player, the  score button.
2. Click  MENU FILES to open the FILE MENU window.
3. Choose  IMPORT MMF and select the file you have just saved.

Check: the score has to appear, with its bars and its notes. The Prelude is in **E minor**, so the key signature — at the start of the staff — has **one sharp**. If you see an empty page, the file you chose was not the .mmf.

And now the surprise — which is not your mistake. What appears does not look like the printed score: **the whole piece has landed in a single channel, CH1**, in the bass clef, with the right hand hanging from ledger lines far above the staff. **CH2** is empty.

The Player did not get it wrong, and you did not save it wrong: **this is how the MIDI file comes**. Look inside it and it has **a single track**, called [Group 2], with all 752 notes mixed into channel 1. The two hands were already merged before you touched anything, and saving it again from the Player would give exactly the same result: nobody can split up what the file never split.

Leave that screen as it is and carry on to PATH B. The comparison *is* the lesson.



7 • It is yours now: play it and look at it

From here on it is not a recording, it is your score:

- **Listen to it** with the Score's transport.
- **Change a note** and listen again: what you edit is what sounds.
- **See it as a page and print it:** that is step 8, with the version that comes from the MusicXML file, which is the one worth printing.
- **Keep it** with  **EXPORT MMF**, so you can open it again whenever you like.

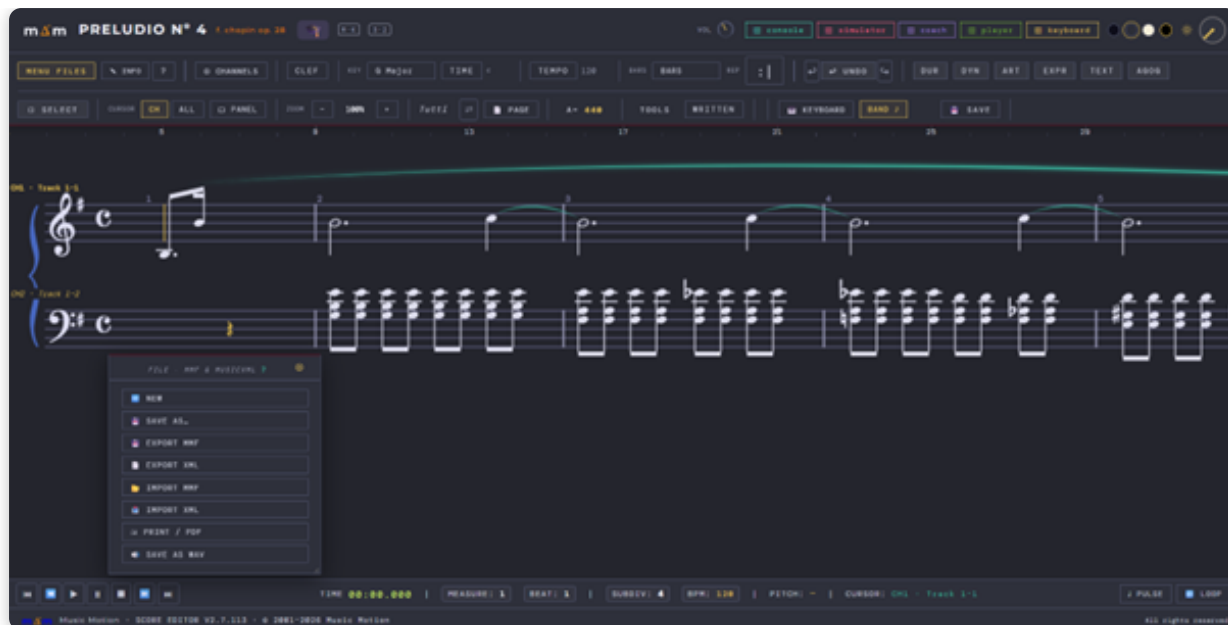
PATH B — AND THE COMPARISON

Now the short path, with the same Prelude, to see the difference:

1. In the Score, **MENU FILES** →  **IMPORT XML**.
2. Choose **Preludio N° 4 Op28 - E min.mxl**.

Check and compare: the first thing jumps out even if you do not read music — now there are **two channels**, one for each hand, each with its own clef: the right hand in the **treble clef** and the left hand in the **bass clef**. It is the same piece you have just seen crammed into a single staff.

And the second, looking closely: find a bar with accidentals — a sharp or a flat on its own, one that is not in the key signature — and look at it in both versions. You have the **PDF** of the Prelude next to the files: that is the original, and it tells you who is right.



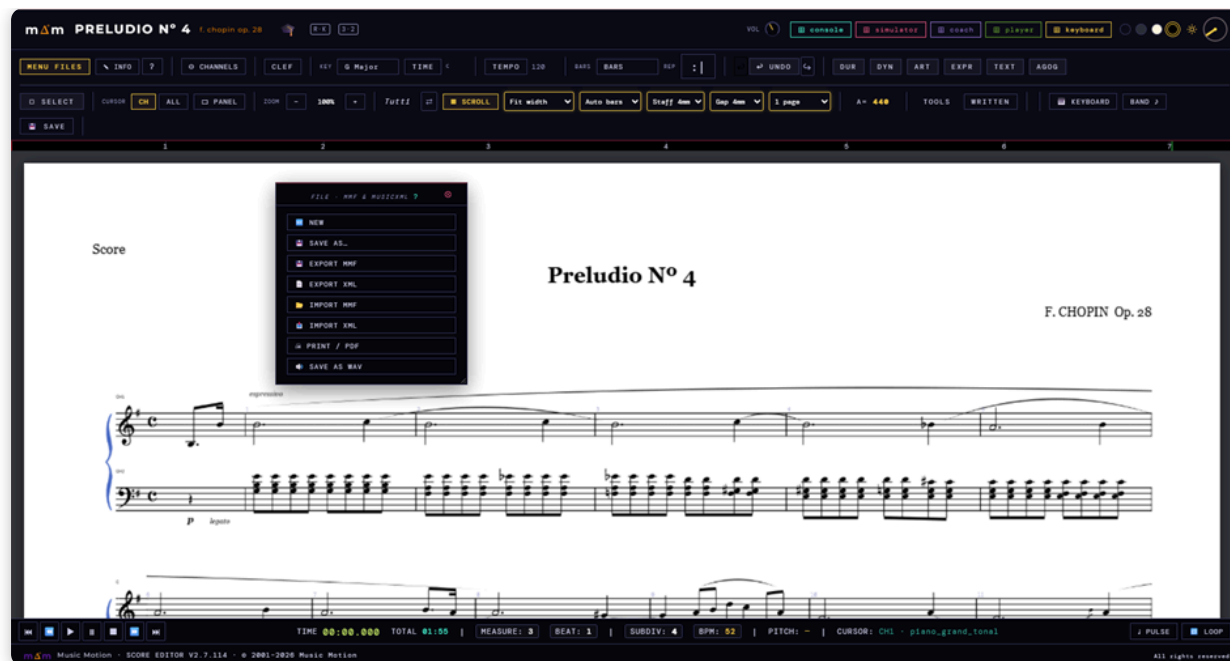
If a note came out written differently in the version that came from the MIDI file, it is not a mistake in the program: **the MIDI file did not carry that information** and the copyist had to work it out from the key signature. The version that came from the MusicXML file did not work it out: it read it.

The rule, in one line: if the MusicXML exists, use the MusicXML. If there is only MIDI, that works too — and now you know what to look at after converting it.

8 • See it as a page and print it

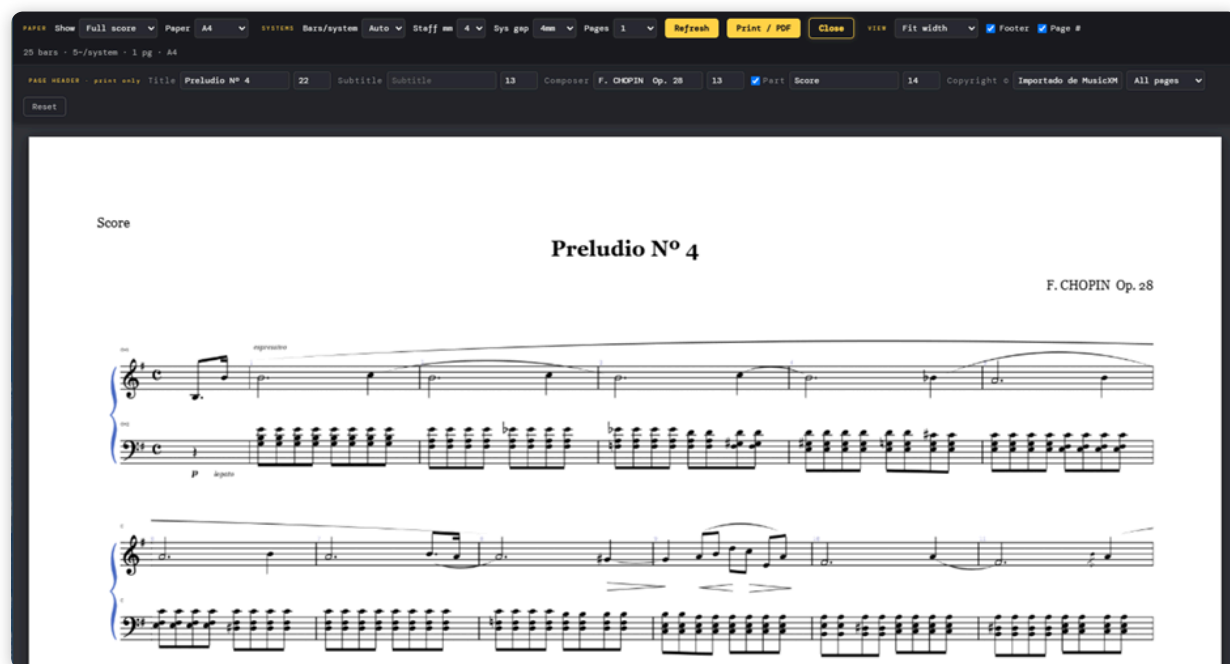
So far you have worked on the **strip**: the score runs from left to right, without breaks. To see it as it would be on paper:

1. Press  **PAGE**, on the second row of the toolbar. The strip turns into sheets: the title, the composer and the systems laid out as in a printed edition.
2. That same row now shows the sheet settings: how it is displayed (**Fit width**), how many bars go in each system (**Auto bars**), the staff size (**Staff 4mm**), the space between systems (**Gap 4mm**) and the pages. You can keep editing on the sheet: what you change there changes in the score.
3. To go back to the strip, press the same button, which now reads  **SCROLL**, or the **Esc** key.



To print it or save it as a PDF:

1. **MENU FILES** → **PRINT / PDF**. The print window opens with the sheet exactly as it will come out.
2. At the top you choose the paper (**A4**) and what is printed (**Full score**: the whole score). Below, if you like, the header: title, subtitle, composer, the part name and the ©. These change only the printed sheet, not the score.
3. If you changed anything, press **Refresh** to bring the sheet up to date.
4. Press **Print / PDF**. Your browser's print dialog opens: choose your printer, or **Save as PDF** to keep the file.
5. **Close** takes you back to the Score.



Check: on the sheet you see the two staves joined by the brace, the title **Preludio N° 4** centred at the top and **F. CHOPIN Op. 28** on the right.

Congratulations: you have converted your first piece
